



SIMATS ENGINEERING

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Course Code: DSA0306

Slot: C

Course Name: NATURAL LANGUAGE PROCESSING

Course Faculty: Kumaragurubaran T

Project Title: DocInsight AI: An Intelligent Document Analysis and Question Answering System

Module Photographs:

DocInsight AI - System Workflow

- 1. Input**
Upload Document (PDF / DOCX / Images)
- 2. Document Processing**
 - OCR (for scanned docs)
 - Text Extraction
 - Text Cleaning
- 3. Document Analysis**
 - Summarization
 - Keyword / Insight Extraction
 - Create Embeddings for Semantic Search
- 4. Store & Retrieve**
 - Store Embeddings in Vector Database
 - Retrieve Relevant Information
- 5. AI Response**
 - Answer Questions using RAG + LLM
 - Translation
 - Text Simplification
- 6. Output to User**
 - Summary
 - Key Insights
 - Answers
 - Translated / Simplified Text

Project Description:

DocInsight AI is an intelligent document analysis system designed to help users efficiently understand, explore, and retrieve information from uploaded documents. The system applies Natural Language Processing (NLP), semantic search, and Retrieval-Augmented Generation (RAG) techniques to identify meaningful relationships between user queries and document content. When a document is uploaded, DocInsight AI extracts and processes its textual content, performs cleaning and normalization, and divides the document into smaller context-preserving sections. These sections are converted into semantic embeddings using transformer-based models and stored in a FAISS vector index. This enables the system to search for information based on the meaning and context of a query rather than relying only on exact keyword matching.

Student Signature

Guide Signature